

Information on the Numerical Analysis module

Instructor Alice Cortinovis, alice.cortinovis@unipi.it.

Office 329A, Computer Science Department.

Office hours (ricevimento): Friday 9–11 or by appointment.

Lectures Alternating with Prof. Antonio Frangioni.

Irregular schedule, especially at the beginning.

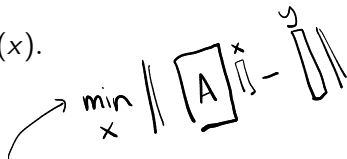
Content

Prof. Frangioni's part: solve all the problems:

$$\min_{x \in S} f(x).$$

This part: solve a very specific problem:

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|^2.$$


$$\min_x \| \boxed{A} x^* - y \|^2$$

(LS)

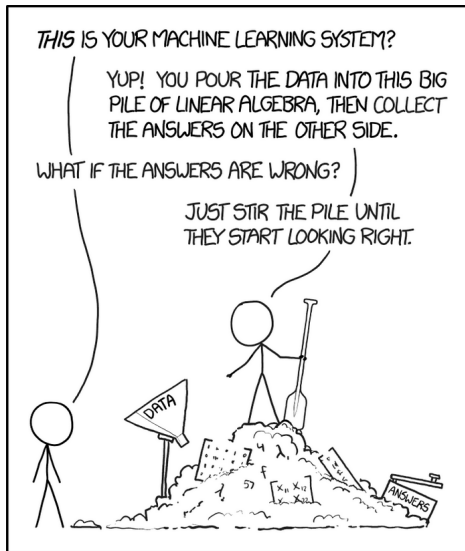
“Solve a linear system, or find the best approximation of the solution in a very specific norm”

Content (cont.)

Why is this still relevant? It is a problem we can understand well:

- ▶ Specialized algorithms that compute solutions quickly and with very high accuracy, even for very large dimensions (think $\approx 1M \times 1M$ matrices);
- ▶ We are very demanding: can we get the solution up to machine precision 10^{-16} ? If no, why?
- ▶ Often appears as sub-step in more difficult problems;
- ▶ Useful for theoretical understanding: e.g., principal components.

Obligatory XKCD



Books

Books (for this part):

- ▶ Trefethen–Bau, *Numerical Linear Algebra*.
Recommended: we will follow its approach.
- ▶ Demmel, *Applied Numerical Linear Algebra*.
Alternative source for other explanations, exercises, ...
- ▶ Strang, *Linear Algebra and Learning from Data*,
<https://epubs.siam.org/doi/book/10.1137/1.9780692196380>.
- ▶ Eldén, *Matrix Methods in Data Mining and Pattern Recognition*,
<https://epubs.siam.org/doi/book/10.1137/1.9780898718867>.

2. and 3. are accessible for free from within our university network (or via VPN
<https://start.unipi.it/en/help-ict/vpn/>).

Languages

- ▶ Matlab: proprietary language specialized in matrix computations. Clunky language with a big Visual Basic-like IDE, great libraries and syntactic sugar for numerics.
- ▶ Python: great simple language with decent numerics libraries stapled onto it.
- ▶ C/C++, Fortran (yes, that's still a thing): sometimes needed for best performance.
- ▶ Julia: newer, same age as Go / Rust. Stable, but tooling still not perfect. It tries to combine all advantages: syntactic sugar for numerics + libraries + fast loops.

Under the core: the same libraries (BLAS/LAPACK) for basic operations: summing vectors, multiplying matrices. . .

How to install Matlab

Available without charge to Unipi students:

- ▶ Go to <https://unipi.it/matlab>;
- ▶ Log on with your @studenti.unipi.it account;
- ▶ Create a Mathworks account;
- ▶ Download and install.

Suggested: install only the toolboxes that you need. For this course: Matlab, Symbolic Toolbox, Optimization Toolbox.

You may also be interested in: Statistics and Machine Learning Toolbox, Neural Network Toolbox.

Alternative: Cloud Matlab <https://matlab.mathworks.com/>.

Matlab introduction: Command window

```
>> a = 19
```

```
a =
```

```
19
```

```
>> b = 5
```

```
b =
```

```
5
```

```
>> a+b
```

```
ans =
```

```
24
```

```
>> c = 2*a + b; % semicolons suppress output
```


Matlab introduction: Accuracy of operations

Default data type: double (IEEE standard binary64).

Warning: operations are only accurate “up to 16 digits”.

```
>> (1/98) * 98 - 1
```

```
ans =
```

```
-1.1102e-16
```

(exponential notation: stands for -1.1102×10^{-16})

Understanding the effect of these approximations will be an important part of this module.

Numerical linear algebra people are very demanding: we have “simple” problems, and we wish to solve them up to 10^{-16} , or something around there. Not 10^{-8} or 10^{-12} .

Recap of linear algebra

Mat-vec product

$$\begin{matrix} & A & & x & & y \\ \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} & \cdot & \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} & = & \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \end{matrix}$$

$$y_i = \sum_j A_{ij} x_j$$

$$y = \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} \cdot x_1 + \begin{bmatrix} A_{12} \\ A_{22} \end{bmatrix} x_2 + \begin{bmatrix} A_{13} \\ A_{23} \end{bmatrix} x_3$$

linear combination of the columns of A

$x_1, x_2, x_3 \rightarrow$ coefficients/coordinates

Def: A basis of a vector space V is an ordered set of vectors

$v_1, v_2, \dots, v_n$ s.t. every vector $y \in V$ can be written as a linear combination of $v_1, \dots, v_n$ in a unique way.

Example $V = \mathbb{R}^3$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow \text{canonical basis}$$

$e_1 \quad e_2 \quad e_3$

$$y \in \mathbb{R}^3 \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = e_1 \cdot y_1 + e_2 \cdot y_2 + e_3 \cdot y_3$$

Q: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \rightarrow \text{also a basis}$

Linear system:

$$\begin{matrix} \in \mathbb{R}^n \text{ (unknown)} \\ A x = y \rightarrow \in \mathbb{R}^m \text{ (known)} \\ \begin{matrix} \text{mxn matrix} \\ \text{known} \end{matrix} \end{matrix}$$

$$A = \left[\begin{array}{c|c|c} a_1 & a_2 & \dots & a_n \end{array} \right]$$

Finding $x_1, \dots, x_n$ such that

$$a_1 \cdot x_1 + a_2 \cdot x_2 + \dots + a_n \cdot x_n = y$$

Example: $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$y = \begin{bmatrix} 4 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

$$A x = y$$

$\in \mathbb{R}^3$

$$x = \begin{bmatrix} 0 \\ 0 \\ 4 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \rightarrow \text{many solutions}$$

$$y = \begin{bmatrix} 4 \\ 4 \\ 1 \\ 0 \end{bmatrix}$$

$$A x = y$$

$$\begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} * \\ * \\ 0 \\ 0 \end{bmatrix}$$

$\rightarrow 0$ solutions

Def: Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say it is invertible if

$Ax = y$ has a unique solution for any choice of y .

inverse of $A \rightsquigarrow A^{-1}$

$$\rightsquigarrow \boxed{x = A^{-1} \cdot y}$$

↓
should use $A \backslash y$

~~$Ax = y$~~
 ~~$\frac{Ax}{A} = \frac{y}{A}$~~ **no**
 ~~$x = y/A$~~

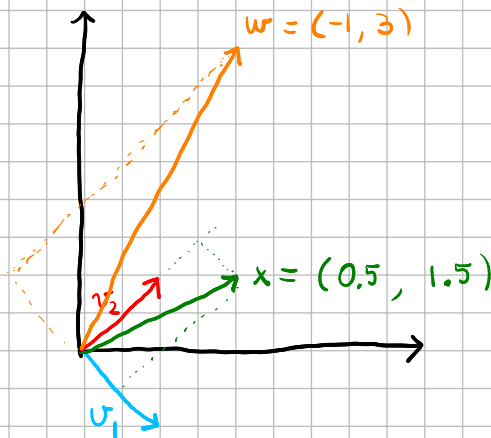
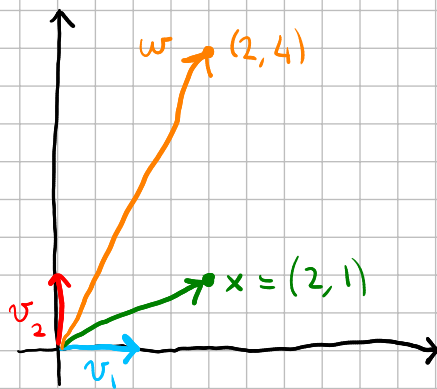
Def: $A \in \mathbb{R}^{m \times n}$

$\text{Im}(A)$ = set of vectors obtained as linear comb. of columns of A
↳ Range

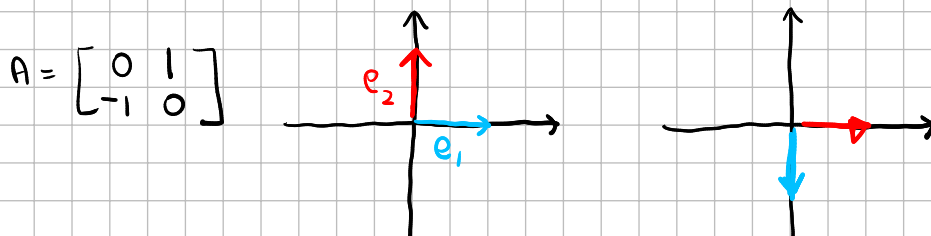
$\text{Ker}(A)$ = set of vectors v s.t. $Av = 0$
↳ Kernel / nullspace

[if you don't have Matlab $\rightarrow$ OCTAVE]

Geometric view of linear algebra



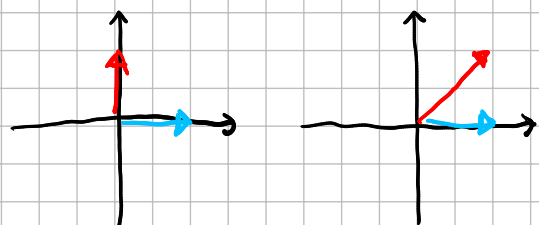
Matrices as linear transformations



$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ $A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$



Q: $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$



$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$I_n = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \text{identity}$

Matrix-matrix multiplication

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}_{m \times n} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \\ B_{31} & B_{32} \end{bmatrix}_{n \times p} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}_{m \times p}$$

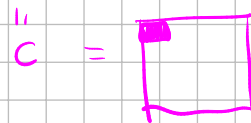
$$C_{ij} = \sum_k A_{ik} B_{kj}$$

Q: what is the cost of



$A \cdot B$

$A, B \text{ } n \times n$



cost $\approx 2n^3$

$A(B+C) = AB+AC$

$(AB)C = A(BC)$

$AB \neq BA$

Q: $A, B, C \text{ } n \times n$

$AB = AC$

$A \neq 0$

Can we conclude that $B=C$?

if A invertible

$\underbrace{A^{-1}}_I AB = \underbrace{A^{-1}}_I AC \rightarrow B=C$

$n=2 \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Def: The rank of $A \in \mathbb{R}^{m \times n}$ is the minimum number of vectors $v_1, \dots, v_r$ s.t. every column of A can be written as a linear combination of these.

"column rank" = "row rank" (thm)

Example:

$$I_n \rightarrow n$$

rank-1 matrix: $u \cdot v^T$

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} = \begin{bmatrix} u_1 v_1 & u_1 v_2 & \dots & u_1 v_n \\ u_2 v_1 & u_2 v_2 & \dots & u_2 v_n \\ \vdots & \vdots & \ddots & \vdots \\ u_m v_1 & u_m v_2 & \dots & u_m v_n \end{bmatrix}$$

Q 3x3 matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

rank-1

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

rank-2

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

rank 3

$$I, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Def. Transpose of $A \in \mathbb{R}^{m \times n} \rightarrow A^T \in \mathbb{R}^{n \times m}$

$$\begin{bmatrix} A_{11} & A_{12} & \dots \\ A_{21} & A_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix} \rightarrow \begin{bmatrix} A_{11} & A_{21} & \dots \\ A_{12} & A_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

"Easy" lin. sys. $A \in \mathbb{R}^{n \times n}$

$$A = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$A \cdot x = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11} x_1 \\ a_{22} x_2 \\ \vdots \\ a_{nn} x_n \end{bmatrix}$$

$$Ax = y \rightarrow x = \begin{bmatrix} y_1/a_{11} \\ y_2/a_{22} \\ \vdots \\ y_n/a_{nn} \end{bmatrix}$$

$O(n)$

$$Ax = \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} a_{11} x_1 \\ a_{21} x_1 + a_{22} x_2 \\ a_{31} x_1 + a_{32} x_2 + a_{33} x_3 \\ \vdots \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \rightarrow \begin{matrix} 1 \\ 3 \\ 5 \\ 7 \dots \end{matrix} \quad O(n^2)$$

Remark:

For general A,

$O(n^3)$

$$x_1 = y_1/a_{11}$$

$$x_2 = (y_2 - a_{21} x_1) / a_{22}$$

"forward substitution"



"backward substitution"